

DIAMOND SHOALS BUOY 41025, USA

WMO: 997040

Lat: 35.025N

Lon: 75.363W

Elev: 5

StdP: 101.26

Time Zone: -5.00 (NAE)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	1.4	-10.1	1.6	0.8	-7.8	1.9	2.9	33.9	13.7	29.0	13.1	11.8	340	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.6	29.6	25.8	28.7	25.2	28.3	24.9	27.2	29.3	26.5	28.3	25.9	27.7	5.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.2	28.9	25.9	21.3	28.2	25.3	20.5	27.6	85.6	29.1	82.6	28.3	80.0	27.7	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.3	21.9	18.0	-2.5	29.6	3.3	0.9	-4.8	30.3	-6.7	30.9	-8.6	31.4	-10.9	32.1		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.8	10.9	11.3	13.2	16.5	20.0	24.5	26.3	26.2	24.7	21.0	16.7	13.8
	DBStd	6.68	5.69	4.74	4.53	3.64	2.89	2.40	1.42	1.44	1.74	3.39	4.34	5.08
	HDD10.0	150	59	38	22	2	0	0	0	0	0	0	5	24
	HDD18.3	948	235	201	167	76	18	1	0	0	0	16	79	155
	CDD10.0	3363	88	74	119	197	312	435	507	503	441	341	206	141
	CDD18.3	1119	6	3	7	21	71	186	248	244	191	98	31	14
	CDH23.3	7609	1	0	0	4	81	1255	2336	2274	1234	371	37	16
	CDH26.7	966	0	0	0	0	1	133	369	371	78	14	0	0
Wind	WSAvg	8.8	10.2	10.7	10.7	10.1	8.7	7.1	7.5	6.4	7.7	7.7	8.9	9.6
Precipitation	PrecAvg	1292	101	87	96	81	93	119	139	157	141	108	80	91
	PrecMax	1809	204	159	184	198	227	194	251	264	269	226	204	169
	PrecMin	734	35	22	29	22	3	21	39	66	17	9	15	33
	PrecStd	204	45	39	40	46	58	48	48	58	68	58	48	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.3	22.6	22.9	23.9	25.6	33.0	33.2	29.8	28.5	27.9	25.3	25.1
		MCWB	20.6	19.9	19.4	20.0	21.4	27.9	27.7	26.5	24.2	24.5	22.3	22.8
	2%	DB	21.9	21.6	22.1	23.0	24.9	28.9	30.1	29.2	27.9	27.0	24.4	23.6
		MCWB	18.6	19.0	18.2	19.4	21.4	25.7	25.3	25.8	24.1	23.8	21.0	20.8
	5%	DB	21.1	20.2	21.1	22.4	24.2	27.8	28.6	28.8	27.6	26.4	23.6	22.4
		MCWB	17.7	17.1	17.4	18.8	20.9	25.0	25.3	25.2	24.0	23.2	20.0	19.1
	10%	DB	19.8	18.9	20.1	21.6	23.7	27.1	28.1	28.4	27.2	25.5	22.8	21.2
		MCWB	16.6	15.7	16.5	18.2	20.5	24.4	25.1	24.8	23.6	21.7	19.1	17.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.9	20.7	20.6	21.7	23.5	28.2	28.4	27.6	25.9	25.6	23.2	23.2
		MCDB	23.1	22.3	22.5	23.2	24.6	32.8	32.4	28.9	27.3	27.3	24.9	25.0
	2%	WB	19.3	19.8	19.1	20.5	22.7	26.7	27.2	26.9	25.4	24.5	22.0	21.6
		MCDB	21.5	21.9	21.3	22.3	24.0	29.3	29.3	28.7	27.1	26.7	23.9	23.5
	5%	WB	18.1	17.9	18.1	19.6	22.1	25.7	26.6	26.3	25.0	23.6	20.9	19.7
		MCDB	20.8	20.0	20.6	21.7	23.5	27.5	28.2	28.4	26.9	26.1	23.3	22.2
	10%	WB	16.6	16.0	16.8	18.8	21.4	24.9	26.1	25.5	24.4	22.8	19.7	18.4
		MCDB	19.4	18.7	19.6	21.2	23.2	26.6	27.7	27.8	26.6	25.5	22.6	21.1
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.1	5.6	4.4	3.4	2.6	2.6	2.2	2.4	3.2	4.6	5.6
		MCDBR	6.3	7.2	6.3	4.6	3.4	3.7	3.5	2.4	2.5	3.3	5.0	6.2
		MCWBR	6.6	7.4	6.0	4.8	3.3	3.0	2.8	2.3	2.5	3.3	5.1	6.1
	5% WB	MCDBR	6.5	7.2	6.6	4.8	3.0	3.9	3.4	2.1	2.5	3.6	5.4	6.5
MCWBR		6.9	7.6	6.6	5.2	3.5	3.2	3.0	2.3	2.6	4.0	6.1	6.8	
Clear-Sky Solar Irradiance	taub	0.341	0.345	0.366	0.407	0.447	0.480	0.503	0.491	0.425	0.376	0.351	0.346	
	taud	2.513	2.502	2.459	2.347	2.274	2.229	2.179	2.211	2.396	2.506	2.527	2.514	
	Ebn at Noon	862	899	906	880	845	813	791	794	835	854	842	831	
	Edh at Noon	83	93	106	124	135	141	147	140	112	92	82	79	
All-Sky Solar Radiation	RadAvg	2.31	3.10	4.17	5.37	6.12	6.38	5.93	5.48	4.54	3.74	2.65	2.12	
	RadStd	0.17	0.34	0.35	0.43	0.43	0.33	0.39	0.37	0.30	0.34	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+174	+117

Nomenclature: See separate page